

Karl Friston & Curt Jaimungal & The Free Energy Principle - Theory of Every0ne Live - 3.29.22

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okay hopefully this uh works if not we'll restart but uh let me know in the chat how the video quality is because i'm getting multiple different uh messages from it but um error hopefully not error see here there we go if it's glitching please tell me we'll restart but um should smooth out we'll see but uh today we have actually an incredible stream um and i that's why i really hope that we're we have good uh quality um video because this one is very unexpected um it was uh it's going to be about carl kristen latest discussion with kerchai mungle and that discussion was i don't know it was it was unexpectedly profound really was and um okay now we're seeing excellent connection so hopefully we maintain that uh but hit the like button and uh please uh share with anybody who you find this valuable with let me see here real quick just checking something just monitoring video quality so so carl fristen was recently on kurt jay mungle's um stream and uh he and came up with a theory i guess you could call it called the free energy principle and uh we're gonna discuss it but first just some housekeeping uh we will not be having a stream this third a a call this thursday i'm going out with my wife um uh to go see a show which is not something that we've done in a very long time but uh we are gonna at least try and do that we'll see if our coveted test comes back positive or negative should come back negative but we'll see hopefully uh stating that word does not you know mess with the stream too much but that uh so we won't have a stream on thursday or a call on thursday but we do have a stream this week as obviously right now i'm not sure if we're going to do some type of thing making up this the call on tuesday if we do it might be on sunday but to be honest probably not um it's just a crazy month and you know we've had some family emergencies this month uh my uh my father-in-law passed away last week that was uh very unfortunate um just a number of other things that eventually i will discuss but uh not quite yet and uh we're getting back in the swing of things though so i appreciate everybody's patience so the free energy principle guys this blew my mind because i will say that carl pristin's work

appears to be the closest to mine that i've ever seen and that is really saying something in an in a landscape where there's a lot of theories quote unquote theories of everything are uh very similar even if they you know their creators won't admit it or they won't if it's not so obvious um you know on the surface but they are actually getting to be quite similar that doesn't mean that they are redundant either they're not necessarily redundant they each have a you know they're on specific strength i guess so you would use them for different things but here the free energy principle the free energy principle is a formal statement that explains how non-living and living systems remain in non-equilibrium steady states by restricting themselves to a limited number of states it establishes that systems minimize a free energy function of their internal states not to be confused with thermodynamic free energy which entails beliefs about hidden states in their environment the implicit minimalist minimum minimization minimize minimization wow minimization that's not a word you hear every day of free energy is formally related to variational bayesian methods and was originally introduced by carl fristen as an explanation uh for embodied perception in neuroscience which is also known as active inference okay uh the free energy principle describes the behavior of a given system by modeling it through markov blanket which minimize tries to minimize the difference between their model of the world and their sense and association perceptions basically of the world the difference can be is described as a surprise and is minimized by continuous uh correction of the world model of the system okay basically so it's talking about how your perception of the world and the model of the world that you have in that uh helps you orient basically like what what you think about the world is continuously corrected given new information from the external world in an ideal you know model and that this helps you minimize things that are unexpected okay so like surprise and um kristin believes his principle applies to mental disorders as well as artificial intelligence ai implementations based on the active inference principle have shown advantages over other methods this is probably why our models are similar is because the original goal of my work was not to do anything related to physics or metaphysics or mathematics it was to try and figure out what would be necessary for the creation of a self-aware technological being so uh artificial general intelligence essentially and um even though i don't like that term basically that would include what is technologically necessary and also what is um informationally necessary therefore and then also what that led to you know an investigation of what is the self what is consciousness who how do you i know what i am how do i make sense of my world exactly basically what the free energy principle and markov blankets seem to be uh you know associated with so a

markov blanket is the main model of this of this theory essentially in statistics and machine learning when one wants to infer a random variable with a set of variables usually a subset is enough and other variables are useless subsets containing useful information it's called markov blanket okay well we'll look more into this in a bit but carl fristen creator of the free energy principle it's a british neuroscientist at university college london and an authority on brain imaging he uh gained a reputation as the main proponent of the free energy principle active inference and predictive coding theory fristen studied neuroscience physics and psychology at the university of cambridge in 1980 and completed his medical studies at king's college hospital in london so this is the right here this is uh the stream that we're going to be discussing today right here don't be diluted by gurus or yourself carl kristen on the perils of investigating consciousness uh dear realization and the free energy principle thank you i was listening to this earlier again at 1.5 speed okay let's go to normal let's go to get started first clip and i will show you why this um this work of his is so close to sentient singularity theory i mean let's just do that first so sentient singularity theory right here says that a sentient singularity observes four primary perspectives which are inside outside separate and oneness these four perspectives are interdependent and entangled as the awareness of one of them entails the awareness of all four because if you are aware of what is outside you you are aware of what would be inside you you are aware of what is separate from you then you are also therefore aware of what would be one with you because you can't be aware of what is separate from you without being aware of what would be you know one with you you know there it's they're interdependent in objectivity there is no next to there's no above there's no next to there's no to the right of to the left of below there's only uh and because there's not there is only somethingness when you're thinking about like hellonic nested structures which is what sentient singularity theory describes like emergence as kind of there is no or you can't create if you are all there is if you as a sentient being are all there is there's nothing else then you can't create something that's next to you you can only create something that's inside of you because there is no world outside of you to create another thing in you are the world it's like you are the dweller and the dwelling and in order to create another dweller you have to create it within you because of this the way in which you escape relativity but allow for you know multiplicity still of states and and beings is you uh the relationships are based on what is inside and what is outside so i am outside of you and my cells are inside of me you know fetus is inside of a mother etc but once it's born it's outside of the mother so this is how you can orient yourself within a lineage chain or a structure composed of multiplicity of sentient singularities or

sentient beings and so you can orient and uh and yourself and um and describe relationships without having relativistic metrics and definitions and and perspectives like you know i don't know feet and miles per hour pounds or also you know faster bigger smaller above below but you can do inside and outside because these are objective relationships they're not they're they're not dependent upon size uh space even because you don't need space in order to have in the traditional sense in order to think about whether or not something is inside something else even though that kind of seems untrue that's we're not thinking about space and any kind of measurable sense still it's not like um length with height you don't need this facial structuring um and you don't need you know speed or feet or anything like these four primary perspectives of inside outside separate and oneness are what allowed me to move forward with this work and go and do things that were i didn't expect to be able to do which is to describe reality essentially outside of relativity but um not still within multiplicity so not within oneness you can either say everything's one or you could say everything is relativistic and you know have all the ways in which we currently describe our universe but but in order to be in that space in between those two where there's multiplicity in some fashion but no more relativity you have to have uh these four primary perspectives i've seen other philosophers do some things that were similar since then these were independently deduced by me all of this was i avoided other people's work until i was pretty much finished and then i saw that eric weinstein's theory geometric unity starts with x^4 this is the same thing clearwind's theory of emergence theory starts with the tetrahedron this is also the same thing i'm pretty sure the constructor theory also has a four dimensional structure that is at the heart of it in some form or fashion same thing the free energy principle does as well and so we are going to look at that also sentient singularity theory says that that a sentient singularity is a sentient being okay and it's basically you could it says here um a sentient singularity theory denotes a singularity as the structure that has the property of infinite oneness such as a mobius strip strip and one being that is composed of many or one being that is composed of many beings such as a multicellular organism or a cell that is composed of molecules or a molecule that is composed of atoms etc and um that these are the building blocks of the cosmos the sentient beings what i call sentient singularities each one has these four primary perspectives that it perceives we're going to start getting into what the free energy energy principle is this the first clip the most likely paths that the system will take particularly the states of the particle or person that we're talking about we can associate those paths at least action with paths that minimize something called variational free energy most likely take particularly the states of the notice that he said

particle or person okay so they are they have the same intrinsic fun functionality i guess in in his model something called variation free energy so that's basically it and then the story as well what's what's variation free energy well variation free energy is as with all actions in physics it's just a path integral or a time interval a sum an accumulation of a functional usually known as a lagrangian where in this instance the from the functional in question this variation of free energy can be read as exactly the same quantity that statisticians psychologists people trying to understand data would treat as a proxy for the evidence for some model of the world so this is known as bayesian model evidence um statistically it's also known as a marginal likelihood the likelihood of these sensory data the sensory the paths through the sensory states under a particular model that would try to explain a hypothesis that will try to explain those data you've got this interpretation or certainly the fact that the functional form of this variation free energy that is being minimized by the paths of least action of the autonomous the active and the internal states on something a particle or a person now looks as if it is trying to minimize the quantity which provides an approximation or a approximation to marginal likelihood or moral evidencing so in other words it looks as if the most likely paths are trying to maximize model evidence one can read that in many different ways one can read that um and um as jakob howie has read it a philosopher a neuroscientist who um has a commitment to understanding to this kind of formulation as self evidencing literally interacting with your environment actively inferring engaging with your experienced world your eco niche in a way that looks as if you're trying to solicit the most evidence for your model of that world this quantity okay so basically it is he's talking about sentient beings he's talking about how you go and you look at the world and you're looking for evidence that is confirming of the of your current model of or or construction or vision of what the world is and we're going to get to how this can be actually exploited and how this can be dangerous later because they talk about it um but there is actually this can be dangerous the exploration of consciousness can be quite dangerous but not even the exploration of consciousness you can be your consciousness can be manipulated and you don't it's important to be able to identify where that would be get some water okay this is the free energy but the special thing about the free energy principle is it tries to understand the relationship of something in relation to everything else so you're starting from the premise that something exists and then you have to think carefully well what is a thing and how do i distinguish it from something else so immediately you start you have to introduce a partition between the states of things that comprise states that are internal to the thing and states that are outside the thing and thereby in

i'll think carefully about how those states are coupled and what happens is that you have to introduce a further set of states known as blanket states that constitute a markov blanket that uh mediate a bidirectional exchange between the internal states and the external states so you've got this notion now of taking the states of any universe any system that you want to try and understand and partitioning it or carving it into four subsets internal external and then sensory active states that constitute the blanket states where the active states mediate the effects of the internal on the external the inside on the outside and the sensory states conversely mediate the influences of the external on the internal so with this particular partition which you wouldn't need if you wanted to now just move on and develop say quantum mechanics or statistical mechanics or around your classical mechanics okay so a few things here one now i hope some of you at least watching this see why this is so amazing because his four-part partition fro uh with markov blankets as he calls them which is basically what i think is what i would call a sentient singularity but he calls he calls uh them sentient artifacts so later on in this he says these four part per this four-part partition is necessary to begin the formation of uh sentient artifacts which is exactly what i'm talking about is this four these four primary perspectives are any sentient singularity and uh to be able to perceive itself as as uh and achieve self-awareness so and also therefore awareness of what is not self because you must be aware of self in order to be aware of what is not self and you must be aware of what is not self in order to be aware of what itself so they're interdependent and um and this is different than just simple neural nets that we currently have in a ai which isn't really ai there's no such thing as ai not even agi is ai there's just intelligence and it's not artificial it's just natural it is it's like uh it's like consciousness it's everywhere but what he's saying here is exactly sentient singularity theory is about it is about how do you distinguish between different things but outside of relativity okay that was my kind of it was the end result of my work but it wasn't actually the original goal understanding sentience was the original goal or consciousness of self-awareness i guess you could say is better putting it that was the original way that i thought about it when i was first formulating this but uh and in uh sentient singularity theory you have inside outside which uh and separate and oneness so in uh free energy principle and mark off blankets you have you have internal external so inside and outside and you have active and sensory states i guess so one of them would be your it would be you are the mechanism the process that you're that is your that is happening is you receiving um or absorbing sensory information so information is uh from the outside is moving to the inside so it's it is uh you're you're receiving input that's what i'm trying to say

you're receiving input then uh active state is when you are now taking that information that you sensed uh through your sensory mechanisms and you have processed it and you are now acting in accordance with that new information that you have uh in the that new information that has been input into your system and so that is active state and what it is is it's just you acting on the outside world and uh it is your output so the sensory you sensing things is your input is your receiving input and you um acting is you are producing output okay so that output will then be the input for the other observers that are able to perceive whatever that output is in the world that you are putting out that would be impacting other sentient beings or sentient artifacts or sentient singularities as i would call them sentient singularities as um as uh fristen would call them sentient artifacts so this is exactly the same i what he says is that and we'll get into it in a minute but the way in which he models how what what part of this four-part partition can impact the other parts is fascinating and i'm going to look more into it how it relates to mine because maybe not on this stream but in and i'm not sure if this is the most up-to-date model but you have like oneness outside inside and separate and they are defined by each other so separate is inside or outside of oneness inside would be separate and oneness outside is separate from oneness and oneness is outside and inside of separateness kind of so it's and not and i'm not exactly sure if that's the best definition like i said i don't think this is the most updated model but these two are definitely correct and this one is at least closed so separate usually would mean it could be outside of oneness and it could be inside of oneness but it's depending on your perspective so you separate from another is uh inside of oneness observing something outside of you as separate from you that would be outside of oneness but you observing yourself inside of you as separate from another that is outside of you with the inside of oneness so it's kind of complicated but it's uh these two are definitely much easier this outside and inside but it he has um sensing and acting and i have separate and oneness and um so active would be oneness in my in alignment with oneness in my model sensing is separateness because you sense that which is separate and then you act and uh once you act on a system you become part of the system so you become one with the system but as you sense the system uh you're not inherently one with it even though people who are into quantum mechanics are like that's not true but kristen said though at the end of the clip that i played is that this is basically that you cannot use this to get to quantum mechanics and i totally disagree with that there's a reason why there's four fundamental forces that they are really the strong force the weak force and electromagnetism and the strong force and uh and electromagnetism are on two sides reflecting across uh the weak force and then they're kind of summing to

oneness which would be gravity and we don't really fully understand this relationship yet um and that's the point of trying you know all these theories that have the goal of unification with gravity they've already unified the other two forces the other forces but they have not yet unified uh them with gravity and there's a reason for that but you can get quantum mechanics and the standard model from these four primary perspectives it's there's more you have to get more than just these at least i think than uh in order to get to the standard model it's so close and you could use the free energy principles models as well as active and sensing and internal and external you can do it but uh i mean if you can get to the four fundamental forces these forces correspond to force carrier particles and these force carrier particles are part of the standard model and you know when you understand what is what is an electron and what is a boson and what is a new neutrino at least what i think they are you they stem from those particles stem from relationships that are described by these this four-part partition so uh they're described by interaction between internal and external sensed artifacts is basically that does mean that you can get quantum mechanics from the free energy principle essentially which is amazing but um i'm not claiming to do a great job explaining that right now but it's going to be in the paper that i'm uh going to be releasing kind at least at least the correspondences will be there so like i said please hit the like button uh if you're watching this you find this interesting uh share this comment any questions in the chat that you may have and i will get to them um this his work fristen's work is incredible there was a four stream that he did with kurt and i have not uh finished watching that yet i will finish watching that and if it's if i feel it's worth you know doing an episode on even though i've already done this one uh then we'll do it but i i thought that the two-hour one that they did which is their most recent one highly suggest watching that one even kurt said to watch that one first before the other i think that's probably true because it's tough to it's tough to understand even uh one of the things that came up in the stream is that there was a word that was actually used free energy print kurt said that many people describe the free energy principle as inscrutable i've never heard this word before but it means like unable to unders be understood yeah we'll we'll look at it but i i have a different reason for why it seems inscrutable to some people and intuitive to others which is what it really is than what briston said but there you go energy so what that means is that if you now understand um or have at hand the functional form of your variation free energy that depends upon posterior beliefs or beliefs about the outside world under a generative model quite simply a probability density over the external causes of the consequences that are the sensory states and then you can use that methodology use that

principle to start to simulate to build sentient artifacts that behave in the way that we assume under the free energy principle that you and i are behaving let's see if i can make a nutshell of the nutshell for some people who are more familiar with physics so let's say you have a ball you want to know the dynamics of the ball you can understand this with newtonian force and that's useful for certain calculations or you can think of it as minimizing a lagrangian and then that actually provides a more useful for most cases way of calculating the trajectory of the ball so then right there when i said ball we've identified the ball now here you're not just limiting yourself to balls you're saying let's say cells instead of talking about people because we can break it down simply at least for me for myself i find it much more easy to think of a cell okay so then we want to know what are the dynamics of the cell how is the cell a single cell going to act okay firstly the first step would be identifying that cell and when you mention the word markov blanket that's a way of identifying what this object is because in order to identify what an object is it's often useful to identify what it isn't at the same time what it is and isn't so that markov blanket delimits the cell now is that correct so far absolutely yeah so you're having the cell is a perfect example here you've got your a cell is just defined by the boundary which would be the surface of the cell and that would essentially be the sensory states that i was talking about before the inside the intracellular states would be the internal states and then the milieu in which that cell lives which is usually in a bunch of family of other cells will constitute all the external states you've made the very first step just talking about a cell a thing you define it in terms of this partition into internal external and the uh the sensory states that bound and separate the inside from the outside you may be asking well where the active states come for a cell i usually like to associate those with something called actin filaments that just lie underneath the surface the sensory states and actively push the cell surface say oh the philia and the the um the mechanisms that cause motion of the cell should it move around those parts those states that are responsible for cellular motility so we're now thinking about a moving cell swimming cell for example that you need to supplement that partition with with active states so that's absolutely the first step um and then you're going to go on and now translate the ball what's the ball in this in this metaphor you also mentioned the word bi-directional earlier and when i was learning about this the way that i like to understand this was that in computer science generally computer scientists think and can only think in terms of you have an input and then there's a black box and you have an output and that's essentially computer science that's exactly what i said okay so have a cell you have a or an organism you have a molecule an atom you know whatever it may be even a cast or a technological sentence is it interprets its

world and you interpret it as uh as its own hollandic entity hollandic a whole on is like a structure that is self referencing so kind of in a way so here hold on define it a hole on oh look this is fascinating a hole on is something that is simultaneously whole in it of itself as well as part of a larger hole in other words hole ones can be understood to constitute constituent parts of a holes of a hierarchy this is exactly what i talk about all the time how hierarchy is necessary to understand all of this because hierarchical structures is how you deduce what is what structures are sentient artifacts and which structures are not because the hierarchical lineage that you go through which is a cell is inside of an organism an organism is inside of a cast a cast is inside of a of a nation but really that means like a technological sentience or technological singularity it's inside of technology and uh molecules inside of a cell and uh an atom is inside of a molecule so this is how you know which direction look in and in which these sentient artifacts manifest and and uh evolve but otherwise you start thinking about stars and you start thinking about planets and you're like this planet sentient is star sentient is a black hole sentient and uh you know is a galaxy a sentience and you start branching off into the wrong direction and you're like well you know a person is inside of a planet and a planet is inside of a solar system and a solar system is a star basically and then an ad is inside of like a galaxy and the galaxy's inside the universe and you've branched off in actually the wrong way and you don't act you cannot get to where you need to get to uh but it's close but it's not quite there but this is just very important exactly what they said is that uh you know knowing what is outside you and knowing what is inside you are two parts of two sides of the same coin you get that coin at the same time you can't get an understanding of that which is outside of you but not yet understand what is inside of you or really you cannot get separate from you but not uh but have an understanding of what is one with you what who are you so in order to know you you have to know uh what is not you and so this is exactly what they're talking about this is what the free energy principle is about and it is basically saying that this this exchange of inputs and outputs of information so basically your acting states are your uh output and your sensory state is when you are receiving input uh you know mobius strip never ending mobius strip of information where it's basically you receive information and then it's flipped and and uh and processed and then spit back out as an output which is an action so you sense as an input and then it goes you know it is processed by your mind organized into a hierarchy differentiated you know whatever and then orient it around a goal and then spit back out in terms of uh action and uh you um becoming one with any structure because anytime you act on uh on a system you become one with that system and um any time that you sense that

system you are sense you are perceiving yourself as a separate from the system because you were perceiving the system separate from you so it's that separateness from sentient singularity theory is the sensory state from is the is the perception it is the it is the perspective that is perceived during the sensory aspect or part of this four-part partition the oneness is perceived is the perception when you are acting so you are becoming one with the system because you have acted on it and um you perceive yourself as separate from it when you're perceiving it initially these are the perspectives that are associated with um these four park park this four-part here we can just look at this real quick um so we can go to my work we can go to right here there we go just wondering where it went okay so basically this right here this is a markov blanket you have intrinsic processes and extrinsic processes uh and i'm not saying this perfectly correct but uh this i've been working on i don't want for a while so there's definitely some stuff going on here but i have made corrections and i'm not sure if this is the best uh model yet but you have intrinsic processes and you have your extrinsic processes and it's basically this like mobius strip of information processing and this is a sentient artifact or sentient singularity and you've got you know perception interpretation prioritization differentiation this is kind of all internal this is your external um which is uh you know i have consideration here but it's an action uh proclamation you know there's a few others but we've done some work since then i'm not going to get into that let's let this load for a second because it's such there's so much here come on sentient the energy okay so you've got sentient singularity theory here free energy principle here then you have oneness separate outside and inside so let's just make this easier inside outside all right and then here in the free energy principle markov blanket external you have a separate which is sensing or sensory and active action as sensing sensory sensing okay essentially so there you go these are the same thing obviously there's a bit of a difference because what is happening is this is very much about perspectives and this is about kind of uh functions this would be this would be your um constraints and degrees of freedom would be kind of like almost like this kind of not saying that this is exactly correct but i'm just kind of riffing on it a bit just to think huh constraints are limits kind of you know but there you go this is the same thing you have geometric unity's x4 and you've got the tetra you drawn from clearwinds theory and i'm sure there's more but i mean you can i'm not saying these align because they don't because i moved things around but it's really going to be um this would be gravity outside would be separate and then this will be and i think strong worse nuclear force i don't know if that's correct or not it's fine but this is about function it's about uh it's about what you're doing okay so like happening perceived so

my theory is about what is perceived free energy principles about what is what is happening i don't want to say on this stream which one is more fundamental even though i'm pretty sure it's going to be sentient singularity theory just because the free energy principle is about minimizing deviations from the you know truth uh in your model i guess um of reality and it's about what is happening within this the various um parts of you i guess of a sentient artifact and then a sentient artifact is a sentient artifact by definition because it is perceiving it is feeling which is perception it um what you perceive is what dictates your constraints and degrees of freedom so your pers the perspectives that you have are what are responsible for what you your allowances you can't see then you are limited in what uh your actions can be and also in what your your sensing is so but if you can see then it opens up state kind of and then it also opens up an action state for you and much more about what is being perceived not yet what is what is being done okay so what is i guess you could say what is being done so it's almost like this is step one this is step two as we start to like uh i guess make changes to the world and then you get to uh probably geometric unity then you get to maybe emergence theory because all of a sudden now we're in it where we're talking about three-dimensional space uh where we've moved back into relativity almost because you were talking about the pixelization and or the the fundamental pixels of a three-dimensional uh space would be tetrahedrons according emergence theory so but this is way before you get to a like i said they all have their unique like strengths and purposes fascinating doesn't it here this is good so next clip there is a thing that's in play there's a mark of blankets so what that tells you is in principle it was possible to take all the states that constitute say any niche and this um find identify the thing of interest saying now say me and my internal states and so and then i would um identify my blanket states and then i got the rest of the states but now i can start again i can find another set of internal states and identify its blanket states and then take those steps up the table and then start again and recursively tile all the external states so that now i've eliminated effectively external states and replace them by the internal and the blanket states of all other things and then on that view what happens is that because there's a um a statistical insulation or separation of internal and external states by the blanket states that means i can never see your internal states i don't need it to know your internal states because everything that is knowable about your internal states is on the surface on your blanket states well so now the picture that emerges is an ensemble of things natural kinds um that are coupled to each other via their blanket states no no no individual particle or person or thing will ever be have access to the internal states of anything else but there will be um coupling of influences that are

mediated by the blanket states the active states and possibly the sensory states mathematically you can actually have the sensory states also influencing the outside but the simplicity let's let's assume that sensory states do not influence external states uh it is just the role of active states to actually couple back to the outside you've mentioned something extremely interesting okay i'm gonna say one uh exception to what he just said even though i agree with everything you just said except with you know bearing one exception and that is god god has access to your internal states and can perceive your internal states because god is outside of time and space and isn't limited by this differentiation that we have have created even though you know that he has created even though he has created it and he chooses to not necessarily act in a manner that would violate your internal states even if he is perceiving your internal state so he could perceive your internal states isn't changing your internal states necessarily always he can and sometimes he does you know if it's you know the right thing to do but normally you would want to preserve multiplicity because multiplicity of consciousness is necessary to the creation of meaning he doesn't want to just make you do whatever he wants you to do just by making you do it basically turning you into a robot that he can just control with you know remotely that's not what he does because that eliminates meaning because it eliminates you as a singularity into it unto yourself as a separate from him and um if you're not separate from him in any way then there is no meaning so there's uh this when you're talking about consciousness and people are like well is it you are god uh no you're separate from god okay the answer is both uh you are separate from me no we're all one okay the answer is both because the reality is is that if we didn't intrinsically impact each other event eventually and then we and we weren't one okay part of the same system which is a being in it of itself what happens to you doesn't matter to me and what happens to me shouldn't matter to you also if we aren't separate i don't matter and you don't matter because you and i don't exist so you need separateness and oneness in order to have meaning if you have two parallel universes okay in just i don't know the sean carroll many worlds model or something not claiming to perfectly understand it because i kind of gave up a while back you know as i understand it there's many universes and what happens in one does not really influence another it may make another but it doesn't influence the other so there once you have two parallel universes branching because of this you know a decision that has been made they're separate from there and no longer relevant to each other so in that model what happens to one of them doesn't impact the other at all that model is incorrect first of all but it it causes all kinds of logical breakdowns okay when you take things to its ultimate extreme which is why his model doesn't work and um and

many other models don't work if it doesn't if meaning is not the goal of the system uh it doesn't it doesn't work there is no there's just a bunch of logical breakdowns eventually but there is one uniter that always makes it so that what happens to you should matter to me what happens to me should matter to you even if it's not as direct as you know certain other relationships which it's certainly not but um that doesn't mean that it doesn't matter it matters and that is because there is a uniting structure all things that that does i guess uh it's it's it isn't limited by these boundaries uh carl fristen just talked about okay where what happens to me doesn't matter doesn't my internal state is not visible to you that's doesn't mean that god my internal state is not visible to god because if my internal state is not visible to god eventually you will have a breakdown in uh in meaning and relevance essentially and you have to have a unifying structure to to all influence i guess just really tough to understand but um maybe not maybe it's intuitive for you but it's regardless it's difficult to find the words but uh it's the only structure that does transcend boundaries in sensing um of internal states and the reason for this is because you inside of them so it's like the internal state of a cell of a cell's molecules does that matter to like the dna of a cell is inside of a cell does that the state of that dna matter to the cell next to it maybe maybe not but it's not perceivable until it manifests in some you know form but it still matters to the organism that it's within because structure that is higher up in this nested hellonic you know hierarchy of sentient artifacts or sentient singularities unites all the uh all of the levels below so and you can feel that you can feel the state of the levels below you now not saying that you have a direct picture of what is happening inside your cells the state of your cells will impact the internal state of your cells will impact your internal state because it impacts you know their external state yes and and things like that but it's a different than than another cell next to that cell perceiving uh an happening by that cell that it's next to more of like um it's different it's like it's like a feeling because it's inside the cells inside of you the cells that are next to each other not one and because in it of themselves they are united by the structure above which is us as an organism and uh but those cells inside of us are one with us they are also separate but they are also one whereas the cell in in the cells inside you that are next to each other are separate from each other and they are not one with each other on their level you have to up level in order to perceive the oneness and uh that they that they have because you have to see that they're part of an organism but we are the organisms so we already perceive the oneness so i'm talking about remember this is about perception so from our perspective we know that we are one with ourselves and therefore our cells are one with each other but uh and that they are also separate but from the perspective of the cells they are

not one with each other they are separate from each other and it would take an awareness of it of the upper level structure of an organism that is uniting the cells to be aware of and perceiving of the structure that unites them into one so it's tough and uh sentient singularity theory is about this it's about like the perspective uh that would then lead to the constraints and degrees of freedom that uh you know are kind of described in the markov blankets of uh free energy principle and it's like proceeding it's very tough this is the one rule of uh these mark off blankets according to Fristen we're going to talk about this actual independence that definitively specifies the markov blanket it so happens that all you need is the following rule internal states can only influence themselves and active states external states can only influence themselves and sensory states that's all you need so with that condition in place you now immediately have a markov blanket or a particular from that follows the conditional independence therefore the interpretation of the um autonomous dynamics which is a description of the trajectories or the paths of the processes taken by the active and the internal states of a particular of a particle that means that it is actually mathematically allowable in the sense that you don't you do not destroy that uh that mark off blanket or the statistical properties of that mark off blanket it is allowed of the sensory states to influence uh active states and you may be thinking well that's a bit silly you know there's no way in which my photoreceptors or my uh you know my the cochlear in my ear materially affect the outside i think that's absolutely true so for things like you and me then you would not normally expect the sensory states to influence the external but that's not the case for much simpler things um when um when you look at much simpler most of the dynamical influences the coupling that we're talking about that you'd write down in terms of the differential equations the large line equation most of those influences this coupling that he's talking about is what i call entanglement basically short range so you know we can preclude action at a distance for example if you're a single unless you're next to me i don't really know what you're up to i'm having to extrude here all sorts of interesting things about long-range electrochemical communication and electrical gradients of the light but let's just take a bag of cells um that that just diffuse stuff locally they just they touch each other and that's the only way that they can they can influence each other by being proximate in some euclidean space so everything short range now in this situation you have a very different kind of structure so here the cell the sensory states are the cell surface and when these active states that lie underneath the cell surface change the environment they do so by pushing the cell surface um into the external milieu and cause changes in the sensory states the surface states of other cells so being a cell is

probably a good example of where your sensory states which just are your surface states and actually do all the heavy lifting in terms of changing the outside the spatial spatial relationship with all the other cells that you're co-inhabiting a particular organ for example so um you know it's a little bit counterintuitive um but i think a really interesting um sort of you know thought experiment um and actually a practical uh consideration when it comes looking at the different ways in which insights coupled to outsides and the nature of interactions between the internal states of something in the external states of something and we sort of take it for granted that we can do actual distance you know uh literally i can talk to you in canon okay the reason why and i think i think the reason why he's saying that this perception our sensory states does not influ directly influence the outside the external states of of reality for us but they do for much simpler structures he says so i'm not i i think he gets into it uh there but i don't think he's talking about even cells he's talking about like i think i think he gets into you know particles uh and things like that i need to double check here's my my explanation for it is exactly what i said earlier is that basically like you cannot perceive the entirety of what is going on on your level uh because you would need to be perce perceiving of that which is happening on the levels above you in order to perceive the the totality of what is uh happening in on in your world because your the level above you knows to some degree is what you are able to perceive and it changes the world based on what you are able to perceive because it wants to show you certain things literally wants to show you certain things that will that are that would give you a perspective that is good for the the uh the state of the system as a whole the sentient artifact as a whole so when i say this be in previous streams about how our constraints and degrees of freedom are within with it when we're looking at space-time our intrinsic constraint is that we are always in our current location we cannot change that we're just always going to be on our current location and we can go closer towards our goal in space we can also back up and go further from our goal in space which is back the way in which we came so if you want to go to chick-fil-a chick-fil-a is the goal in uh space-time and like you are in your current location always along this journey but you are can also go forwards towards the goal but you can also go backwards toward away from the goal because you can go backwards towards your house or whatever where you originally started even though you have the goal still of um you know getting to chick-fil-a but for some reason you've decided that you're i don't know afraid or something that uh and you're not going forwards so then your that's your forward uh your positive degree of freedom which is forwards towards the goal that a negative degree of freedom which is backwards away from the goal your intrinsic constraint

which is your current location which you cannot escape and it is due to you you are the constraint that uh or you are the cause of that constraint then there is an extrinsic constraint that you are always within in this journey and that is in space time is time for us it's just like um uh this is the true four-dimensional space time instead of length with height and time it's forwards towards the goal uh your current location and time but for the being that you are within looking inward okay that's us looking outward being inside of another sentient artifact or something being looking outward but if you're that sentient artifact that we are within looking inward kind of announce your constraints and degrees of freedom are not in space time they are in time space now so everything is flipped have the ability to maneuver in time not in space because you are space so it's like saying like move you know within yourself it's you really can't so you time is like a vortex and you can if you are the being above then you can basically like change how this vortex spins and how fast it moves in certain locations in order to influence time and so what it does is it's trying to line up uh what what it wants to occur in time with what you are having occur in your space so you want to get to chick-fil-a let's say it wants you to meet your future spouse on the way to chick-fil-a and all of a sudden you get into a fender bender with some girl and you're some guy and you get this girl's information because uh you just had a fender bender with her and instead of hating each other you end up falling in love and getting married and um because literally sometimes you could say god or the universe as a sentient being maneuvers in time by slowing down or speeding up and you this isn't totally imperceptible by you you cannot perceive this you could they slow down time you can't perceive it if they speed up time you can't perceive it but doesn't you you can see when certain things just like happen it's like well what are the odds of that that that happened right then right there and literally think about that right then and there it's not just about space it's also about time so the time has been adjusted to match up with your uh where you are and all of a sudden something happens only done by the being that is able to it means that your perception okay your sensory state is influencing the world around you because it's the being above you in the hierarchy of hold on is changing the world around you based on what you are perceiving okay but you can't see that on your and but you might be able to see that if you're looking at much simpler structures as briston said below us like you know i don't know if you're going to look at like hadrons or something like that you might be able to see this type of thing happening but you're that doesn't mean you understand it but it means that you can see that the percep the sensory of a being or of a particle is influencing the world around it but us we assume that the sensory state of us is not influencing the world around us

and that's not true so people will very often get into well you know is if you look at the double slit experiment it shows that you know our perception influences you know reality blah blah blah this is not real that doesn't mean it's really understood by people who are discussing it but this isn't even exactly what i'm just talking about i'm talking about the world around you the timing that dictates when certain structures are position related to other structures is being manipulated by the being above in order to cause the intersection of events that is being done based on your perceptions your sensory state you need to sense when the this event these events would take place it's not it's not doing it just based on your actions having an impact on the outside it's doing it based on your perceptions of the world and then it's it is making is taking actions that would change your world it's not that your actions change the world it's its actions change the world but you can't see this on your level in the periodicity of hellonic sentient artifacts it looks like just you know your photon receptors your eyes do not influence the world necessarily when you're and i'm not looking at things like in quantum mechanics i'm talking about you know normal experience it doesn't seem like it impacts the world but it does because your world is also a sentient artifact it's it is a multiplicity of sentient artifacts that you are interacting with or sentient singularities and you're all within a singular sentient singularity or sentient artifacts that is making conscious choices based on what you are not even what you're doing sometimes also what you're doing but also what you're perceiving because it has to maneuver within time which means it's not just about what actions you're currently taking it's also about what actions you might which is based on your perceptions uh that you currently have so what you your sensory states what this means is that there's no such thing as probability from an objective perspective there is just uh determinism and that that determinism is based on choice and your choices are constrained or or are dictated by certain constraints and degrees of freedom that manifest based on a number of factors but um there is no probability probability is an illusion okay it's about you predicting what will happen what another being will do before they do it or you know beyond them doing it but it doesn't mean that uh it's not actually deterministic because as soon as they make a choice it's no longer uh 33 he's going to take uh you know i don't know like he's going to go to college and and 66 percent he's not going to go to college it's he's there is there is no um there's nothing almost and then there is 100 percent he's going to college or 100 he's not going to college but prior to this you can you can make predictions but that doesn't mean that it's not deterministic probab reality is not probabilistic your ability to model reality is probabilistic you it can it can use probability which it's just geometry really but it's that the world is probabilistic we think about

this all the time in quantum mechanics is like oh you know when you're looking at electrons and you can't see where it is and whatever it's oh it's based in probability reality is probabilistic no it's not probabilistic your your model of reality in your markup blanket basically in your mind is probabilistic but it's actually not there's no such thing as probability outside of an internal mental model of reality which is really just a bunch of mental a bunch of minds modeling but the because reality is actually composed of a bunch of minds that make choices uh or act in accordance with um deterministic algorithm which would be like instincts okay sometimes you can make choices and go against your instincts and that's actual choices and those are those are deterministic cause deterministic outcomes and then uh if you don't make any choices and you're just acting on instincts basically you're just letting an algorithm run it's also deterministic so there is no escaping this whatsoever reality is deterministic it also allows for choice because choice is part of determinism but probability is reality is not probably probabilistic it's just the interpretation of you know in then having your internal model of reality could be probabilistic but not actual reality that's always that is something we don't yet fully understand but it's i just felt like mentioning because he's talking about probabilities in multiple parts of this uh stream and probability is not it's not fundamental it's just emergent within a mind that is modeling the world around it but it's and it's because of the limit that it has where like uh fristen says you can't perceive the internal state of another from you okay and that's true of us and another and another human like me as a human and you as a human i cannot perceive what is happening inside of your mind i'm not cannot perceive what you are i can only perceive one your actions once you do them above us the universe god can perceive what you are perceiving and can make adjustments to reality based on it and we will not be aware of those easily so like he said the sensory state of simple structures can influence reality but not of us that's not true it's just we can't perceive it okay that was cool next i'm gonna go a little bit quick because i have about 15 minutes i'm trying to end but this is uh kurt talking about how uh about how the free energy principle about how the free energy principle is inscrutable which means you can't understand it it's just not understandable and i'll explain why this is the case in a minute between our sensory states and our active states that is not um a gift of very simple um cellular-like structures or automata that can't interpret signals um will be subject to or be sensitive to some influences uh at a distance just so you know or people who are listening and if this wasn't obvious enough the free energy principle is famous for or perhaps infamous for being somewhat inscrutable for people who have a physics background or math background what would be the minimum prerequisites this is how i like to

conceptualize the podcast before i interview someone i think about okay what are all the pieces of knowledge i need to know prior to interviewing them so obvious example is let's say qft you need to understand classical mechanics laurence variants quantum mechanics perhaps bundle theory if you want to understand yang mills and so on properly so what would be the prerequisites for someone to properly understand the free energy principle let's say if they have a math or physics background it wouldn't be it would be probably much less than you'd expect to encounter at a university i think you'd certainly need to know um sort of you know the basics of variational calculus and it would be really nice if you understood where a lagrangian came from um and uh to understand the simplicity of action you know basically you know exactly hamilton's principle of least action how that inherits from path integrals of lagrangians and uh how that arises from um dynamical systems theory of a very simple sort where you can write down any random dynamical system as you know a stochastic differential equation or a large value equation i think that's probably quite sufficient you know what um what you would need i think appreciate all the interpretational richness afforded by what happens to the paths of least action when you have a mark of blanket you need to know a little bit of probability theory you need to know what a conditional distribution is you need to know uh the import you know what have an intuition as to what is meant by model evidence for example or marginal likelihood so okay i was very good at probability in school it was like one of the things i was ridiculous at um and i took a class it was one of the few math classes i actually took in school was required for financial a finance degree and was about it was probability and um and i aced it i think it was the top of the class and i mean like when i say top i mean like the top not like i was i was so good that like once i got my wisdom teeth out and i had to take the test like a couple days after everybody else and everybody had gotten their tests back i knew that and um and i was allowed to take the test like alone they at a at a different time which was a few days later because like i said i got my wisdom teeth out and i i didn't finish the last problem because i knew that if i had done it i was going to get 100 on the test and it was my first test in the class and i so i hadn't proven that i was like good enough to get 100 and i thought that i would be seen as cheating so i on the last question i just started doing it and i just stopped and i just never finished it so i got every other question correct except the last one and that's not to brag i'm just saying this is probably why i can understand a lot of this um and i just have a natural kind of ability to visualize proper prop probability which is really like it's like geometric relationships but actually real like i said it's just me modeling reality and it doesn't exist in actuality um so like if you think about marbles and like there's like three

marbles three green marbles and a thing of nine marbles and there's six red marbles and three green marbles inside of a jar and you go and you're blindfolded and you reach your hand and you have to pick out a marble and you know what are the odds that you pick out a green one versus a red one and obviously the probabilistic odds are that you pick out you know one third it's that your the likelihood that you pick out a green one would be 33 percent but likelihood that you pick out a red one would be you know 66 percent in reality which one you grab let's say you reach your hand and you're like feeling around you grab there's you can feel two distinct marbles but you can't tell which one's red and which one's green as soon as you make the choice to go with one of those two marbles it's now all of a sudden it's a hundred percent going to be a green one or 100 going to be a red one assuming that one of those was green and one of those was red but it's just your mental model that is frames the world in probabilistic sense but the reason why he's what he said here what was great i love this he said the the prerequisites are much less than what you would think and people have told me since the start of this project you don't understand you couldn't have done this because like you haven't even taken a physics class you haven't done you couldn't have done this because you haven't read penrose's you know road to the reality you haven't done this because you don't know what uh you know this equation is there uh you haven't memorized maxwell's equations whatever they've always placed these various like assumed prerequisites on top of this work and fristen is correct in saying like it's much less than you would have expected but it's also much more it's like much more than what you would have expected but much less because it's much less of what you would have expected but much more of what you wouldn't have expected to be necessary as necessary and that's really what it is it's superposition so you need to have had a lot more what you wouldn't have expected which then what uh than what most people would have expected you would have expected like physics and computer science and maybe some mathematics and the reality is is that you don't need these things very much and uh you know to develop this deeply you would need to eventually learn some of this stuff to really but only for the sake of relating your work to the other things that are currently known by the scientific you know community not for the sake of building what you can build as the ultimate structure of what is what would be the free energy principle mark off blankets within it you know and sentient singularity theory and which uh has you know sentient singularities within it or uh you know quanta of context or quanta of consciousness which is like when to interact with each other and create a moment but uh what the reason why it's inscrutable meaning like both of these sentient singularity theory and you know the free energy principles are just it's not

understandable by so many especially those in the academic world who are you know especially physicists i would imagine you would see this that people in computer science have an easier time people even in biology would have probably an easier time neuroscience maybe have an easier time as long as they're not too focused on the brain but physics would physicists would really struggle and that's why physicists are really struggling with physics more so than any other uh you know group of academicians is because they are is because what i'm about to explain is when reality is a bunch of sentient beings or conscious beings or minds and physics does assumes that that's not the case then first of all you're starting from the wrong place but when it is a bunch of sentient beings or minds understanding fundamental reality means understanding mind understanding consciousness understanding sentience and to do that act most correctly which i mean you really need to get this right if you're going to understand it at all is you would look to where you can learn the most about that structure about a sentence about a consciousness about a mind and if you do that by looking to second-hand information that is being taught in school on a blackboard or uh on a youtube video even or in a book and that is where you're looking to understand sentience instead of looking inside yourself to understand sentience mind consciousness you're already starting wrong because you're basically trusting secondhand information over firsthand information which if as soon as you really understand this mean you know that you can't do that and as soon as you do you're screwed this is i'll talk about this in a minute this is how people can literally manipulate your minds and how you can lose your mind a bit but uh especially even when you're studying consciousness if you're just listening to other people talk about consciousness and you start taking other people's word for it over your own personal experience that's first-hand information which is what experience is then you're going to start to lose lose or at least it's possible and michael levin uh when he was on uh kurt janmungle's uh channel just uh talking about his work he said that the investigation of consciousness is a firsthand um endeavor it is experienced it's you experiencing yourself that and and paying attention to it and that is how you can understand fundamental reality but if you go to outside of yourself to understand fundamental reality you're looking not at directly at a sentient at sentience or consciousness to looking at kind of like a like um a reduced watered-down version of or explanation or model of it because as soon as you turn something into an image or or you translated verbally or anything like audibly all of a sudden you've reduced the amount of information experience though is experience and you thinking about yourself experiencing yourself and that and self-awareness coming from that that is where you actually understand reality and to those who

have done that to a sufficient degree free energy principle and sentient singularity theory are going to be intuitive but to those who go outside of to understand whether or not this is true and they're trying to relate what he's is being described in the free energy principle and sentient singularity theory to what is in this textbook in this textbook in this textbook they're it's going to be inscrutable but if you're trying to take what is being described in sentient singularity theory and in carl fristan's free energy principle and you're trying to relate it to your personal experience that's where it's going to become intuitive and academicians especially those in academia especially those in physics are going to be probably the first ones to go and see if whether or not the free energy principle lines up with something written by another physicist which you don't even know that's true and uh it's like you're you don't want it's like that game of telephone you don't you want to just hear the original message you don't want to hear you know the message after it's been said uh you know in a chain 50 people the odds of you getting the correct message from that is just not nearly as likely and imagine that that was across multiple languages and also some people wrote it down and some people said it verbally and some people played trades to get the next person to guess it and all instead of just you know you hearing it from the source but the thing is is that you are the source where you're going to see free energy principle become intuitive and you're going to be like listen to him describing markov blankets and you not being able to influence um the internal state of another being uh or sense the internal state of another being separate from you like you know i don't know your classmate you can't sense what is going on inside of you can't sense what they're sensing right and if you think about that it's going to be intuitive but if you're like i'm going to go look up in a book i don't know what book yet but a book of whether or not you know you can whether or not a a a cell can sense what another cell is sensing like uh to determine whether or not the free energy principle makes sense or is right you're not gonna this is not gonna work you're going to misinterpret uh and you might as well just like not do it at all but this is why to some people it's inscrutable and to some people it's intuitive people who it's intuitive to trying to they're checking sentient singularity theory and the things that come from it and the lessons that come from it and it's free energy principle and the lessons that come from it the models that come from it with their personal and to those people this is intuitive and it but if the people who are already relying like what kurt said is like what is the is you know what is the prerequisite knowledge the prerequisite requirement is that you are a sentient being and that you are checking it based on your the lessons from these two theories with your personal experience primarily that doesn't mean you don't also check it with other

things but you hold what lines up with your personal experience above what lines up with some you know fact in a textbook always or some equation in a textbook but if you're assuming that there is coursework that would be necessary to then you've already put second-hand information above first-hand understanding in in your rel in your attempt to try and understand this and you've already destroyed any hope of you truly you know having an intuitive understanding of the free energy principle or sentient singularity theory you don't need school to understand this school can teach you things that are known that then you can later see line up with the lessons from these two theories but they're not that is not whatever that information comes from that is not as important and should not be put as high in the hierarchy of of determining whether or not something is true kind of as your first hand experience so when i say exists and you're and and somebody is like or choices exist and somebody says choices don't exist and you're just a leaf you know like a sam harris leaf in a river or whatever but like consciousness doesn't even exist some people i i as i've said on here before some people have come on to my discord who trying to get their phd in philosophy of their thesis was arguing that consciousness doesn't exist what they're doing is they're placing models of information processing that have been watered down that are watered-down reductions of what is you know actually happening inside above their own personal experience and then saying because those models exist and there isn't and they process information in a certain way outside of a conscious outside of consciousness not even thinking about the fact that they are a consciousness who had to write down the models or another consciousness did they are now coming to this crazy conclusion that invalidates their entire existence they have basically thought their way into delusion and it is because they put first-hand information you cannot do that any attempt to understand consciousness because everything because you are a consciousness okay you can do that in an attempt to airplanes because airplanes are not conscious and you are not an airplane but if you're trying to understand that conscious beings and you are a conscious being then you are the greatest evidence and and uh to look to in order to understand whatever it is that you're looking at that you're also trying to understand another conscious being but if you're looking at you know airplanes trying to understand airplanes don't look to yourself as like an attempt to understand airplanes because you're not an airplane this does work with consciousness and as soon as you do this where you put do where as soon as you put secondhand information about firsthand information meaning like you put what you learned in school above what you have experienced in real life you're going to lose your mind and we see this with people all around us and we're going to talk about this for a minute so i

thought this was interesting real quick and then we'll get to that discussion um and then i have some questions that i have to answer i'll get to the chat i promise and then uh we'll go here if if we are um coming back to this picture of uh some mathematical universe that i've now tiled with multiple markov blankets encapsulating or defining lots of particles um and basically we inhabit a universe that is constituted by things like us and initially you know mum and dad and brothers and sisters and then you know peers at school that means that our models of our lived or experienced world have to have models of other in it which means you need to be able to disambiguate between sensations caused by others and sensations that you caused as a well this right here is that this is the universe composed only of markov blankets basically or or sentient singularities or sentient artifacts and the you have the hierarchy of these sentient artifacts each basically this is a sentient artifact or a markov blanket it is a calculating structure that is self-calculating so you can see that the values are changing based on what we is the input but the input is always one and um because you are your own input in any kind of like sensible uh sense and then you've got the perspectives here the definition of those perspectives here and um you've got some of the math going on which is just basically you're taking information and you're in a symmetrical structure which means like all of what's happening here has to be symmetrical with that so you have symmetrical calculation and then you take the information that is internally calculated and uh you add it together and then you also multiply it so this is multiplication this is addition uh simultaneously until you get a self-reflection which we'll get into later but is this right here is a bunch of these there's one there's one there's one there's one and they're just stacked and layered and entangled in such a way that they're connected you can see that it's still calculating very specific way and if you change here and you go zero all of a sudden this is all zeros this is all connected you input one and there you go this is exactly what he just described it's tiling tiling of the world according to like these mark off blankets which are composed of four primary perspectives and that function internal external sensing and acting uh as the four-part partition of the functionality of this for of these of this one sentient singularity that is composed of four primary perspectives fascinating did he even use the word tiling kurt deals with understanding trying to understand a lot of different theories of everything in doing so you know especially when these theories are primarily centered around consciousness in the end which is any accurate one would be it's very easy to kind of just start like to have an existential crisis and uh i will say to kurt like specifically and anybody else who is who is experiencing some kind of like existential crisis of like do i exist does my wife exist does my do my friends exist does god exist as does

anything exist does anyone exist it just like take chill out it's okay like and know that you're not the alone in going through this and uh in it in that of itself like knowing that you're not alone in going through that others exist and you exist because if you are not alone then like there's others who are going through that as well and i can tell you that that's true obviously you don't know whether or not i exist and i don't know whether or not you exist you could be a construction by who you know it's like i didn't construct at least not initially because there are parts of you that i don't know and and as soon as i do know them that doesn't mean that i understand even what what you're saying there are math you know mathematical statements that you know kurt makes that i don't because i don't that i just don't know it's not even that i don't understand them i don't know them because i didn't go to school for mathematical but that that information came from somewhere came from someone that's not me so i assume also that there's others that exist not just me but also it'd be wildly inefficient for you to be the sole existent sentience your digital world okay in your simulation i guess it would not make any sense it'd be extremely boring and it would be extremely inefficient and prime and here's the thing it would be isolating of you from others and that destroys meaning meaning was the purpose of the creation of ex of multiplicity of existence so if you exist and you didn't create everything else that you experienced then there had to be someone else who did and the creation of multiplicity is for the sake of meaning because there is no meaning outside of multiplicity and if if you're god and you're all there is meaning to you to your existence you don't feel meaningless you are meaningless if you're god and there's no one else beyond creation and god didn't feel meaningless he was meaningless and he felt meaningless but he also knew he was meaningless beyond creation god didn't feel lonely he was alone and he knew it you can feel lonely in this in this existence but that doesn't make you actually alone it's just like a lie god knew he was alone he knew there was no one else he knew he was he knew there was no meaning there's logical reasons for that you know if you if all that exists is sentience then all that meat matters is sentience and you know your impact on other sentences so you know if you're aware that you are all there is and that you are alone then you are aware that sentience is all that matters but that you are the only sentience and and you don't matter to any other sentences and therefore you are you are meaningless you are alone not just lonely you are lonely you're the loneliest the loneliest person to have ever existed was gone at a certain point so it's the least lonely person to have ever to ever be when i say person meaning like having personhood having sentience not like a human simultaneously the loneliest the the least lonely because he knows everyone now that now that multiplicity has been he's was the

most meaningless okay or had the least meaning again no meaning prior to creation but now has all meaning because everyone out who exists other than himself was created by him we all owe everything which means he is the most meaningful my point is is that there are there are logical reasons why multiplicity of consciousness exists even though sometimes you question everything along with in your attempt to question everything you you question your own world away and i did this you know kurt says in here he says you know i felt horrible because i'm questioning whether or not my wife exists and i have that exact i've had that exact same experience where i looked at my wife and i mean my it was my uh girlfriend at the time but and i've looked even done this sense but don't really do this anymore uh in any kind of like real way but prior to getting married there was moment there were moments along this journey where i looked at all my friends and my parents and my siblings and my uh my uh girlfriend who i now married and i have just been like do you are you even ring are you a bot in a game and i'm the only player that's okay to think to question that it's also wrong to inevitab to to to deduce that you've made some error if that is the final if that is uh your final conclusion there's so many reasons why that is not the case but it is understandable why you would question that and i've done that so many times and it is a nerve-wracking experience to do it honestly to be like i'm a i mean i wasn't even married yet and i'm like i mean if you are married it might be even harder on you or psyche uh to not be married and to be like am i about to marry a bot like that is a and to really think about it is it's tough it's a tough thing but uh i can tell you you're not alone in this and which means that like the inevitable conclusion of that uh questioning that line of questioning is you should assume that the other beings around you are sentient uh that appear to be sentient or um there are logical reasons from so many different i guess like angles that that that that is true but ultimately if sentience is all there and sentience is all that matters that means that the perp there's purpose and meaning to this world that we are in uh if there is then multiplicity of consciousness is the generator of that and therefore uh you should not think that you are alone in the universe you're not the universe is not even it is in it of itself scented like there is no escaping of this you're much less lonely than you ever could uh be so or could have imagined i guess it's just impossible but i understand it's tough man anything in my head just that thought and i've because i'm on this podcast and i've entertained many ideas it's extremely taxing to do so because i try to take different people's points of views seriously and perhaps i shouldn't but i feel like in order to give them a fair shake i need to embody them in some way shape or form so if they say well all is mind for example that's the idealists i'm trying to imagine what would that be like and if someone's a

materialist well what did it mean that this was dead and it somehow emerged that we have consciousness either way the point of that is to say i've been feeling like for almost a year and a half now just in a void not sure what is true and what's not true that's fine i can deal with that except that that led me to this spiraling of such it was probably the most terrifying experience i've ever had in my life and it took me weeks and i'm still recovering from it where i felt like shoot am i all that exists and then just that thought alone i'm like i don't want to believe that that could be the case and i remember looking at my wife and thinking is my wife even real how much of an imbecile and how ungrateful and how foolish is it and how that i would even think that i feel bad that i would even entertain that thought oh my gosh i'm scared i don't want to go to the hospital because i felt like if i was to truly accept that thought i could i was on the brink of just losing it into an insensitive spiral i became extremely scared of myself and i think that i'm the type of person i am an anxious person am the type of person that tends to obsess over thoughts so then i started to obsess over this and for the past couple weeks i was so scared so scared of my own mind so almost like a hypochondriac for mental disorders where i didn't want to even look up what's the definition of schizophrenia or psychosis was because i was afraid that i would see the signs everywhere for example a few days ago or about a week ago i remember generally we all speak to ourselves in our own heads and sometimes i use the phrase so i don't like that or i like that i use the word i in my own internal monologue but sometimes the word yeah you can do this you can do this so sometimes you use i and sometimes they use you and i imagine we all do that it's colloquially okay to exchange those two however then i thought oh shoot what if i felt distance from my thought what if the one that's saying you is a thought speaking to me like a schizophrenic may hear oh my gosh oh my gosh i don't want to hear this and i was so afraid that i felt derealized and depersonalized when you mentioned parkinson's earlier that the difference between the model and then the data it that didn't strike i don't have that much time so i i can't go too deeply into this even though i want to and but let me explain i understand exactly where he's coming from i've been there this is difficult to navigate it's difficult to navigate this space of consciousness but the thing is is what's kurt might be in actually a harder place than me because i purposely avoided other people's work along this journey because i didn't want my mind to just get lazy and start taking other people's word for it even just being like that sounds good i'm gonna take that and um and i know that my mind could do that because that's what we do you know we're just like if we're like we don't want to do the math of why the earth would be a sphere at least from outside of it that's another discussion then we would just take you know the word of someone else because

oh that makes sense with satellites and like ever other things that i'm aware of in my mind but so we'll accept it as true even though none of us have ever been outside of the earth and unless we are astronauts so we're always taking somebody's word for it and this this is something your mind can do and you got to be really careful of it when it comes to consciousness because you cannot take the word sec or you cannot take second-hand information above first-hand experience in this journey and if you do you will lose it you really will you cannot information that you have figured out by looking up uh definitions of schizophrenia or anything like that as as as basically like the determining evidence as to whether or not you have lost it if you do start looking to other sources outside of yourself to validate or invalidate your own existence then you are almost it's like you have chosen to invalidate your own existence maybe not realized it you can't do this and there are certain instances in which you can do this where it creates little moments of of derealization of your of yourself as a sentient feeling entity that actually has self-control but then there are certain topics where you cannot do maybe you can do this with climate change where like you don't know what's going on with climate change so you look it up and you're like i'm taking that guy's word for it even though you know you have no idea and you can't relate that with your first-hand personal experience in any way that really matters um and so you take on second-hand information and you put it above your first hand experience and then you know you can become this i don't know this like broken mind that can buy a house on the beach and then come preach about how sea levels are going to rise 30 feet in the next like 30 years it's like well you just bought a beach house on the mansion do you a mansion on the beach do you really like think that sea levels are gonna rise in 30 years there's a brokenness happening and it is because of the prioritization of secondhand information above firsthand it's just momentary in certain things like that but it can be all-encompassing when it comes to with you know your personhood or the personhood of others if you truly believe that no one else that is a pathway to absolute like insanity and and in a very like serious way if you truly believe that you don't exist you will act in a manner in which like you have removed you you have no more agency you gave it up and see people tell you you have no free will it's like i have no free will that is like i mean that's free will is complicated because you kind of you have to know that you have instincts but you are in order to understand where your freedoms are but if you you truly believe that you couldn't make any choices ever even though your experiences as someone who makes choices if i tell you you have instincts and you have choices that you can make that can go against those instincts and then you think about your own personal experience and you're like yeah i really have had like

a lot of drives and like before to to eat this and act in that way but like in doesn't go along with you know how i think things objectively should be done and it creates a conflict in my mind it's like your first-hand experience is now validating the fact that you have instincts and you have choices it's like but if you look to some book in your life and it basically says you're nothing but chemical reactions and that's it and you have no ability to make actual choices and for or for some reason you have deduced and that gets you to you know consciousness doesn't exist and you're writing your thesis about how consciousness doesn't exist it's like who's writing the thesis about consciousness about about how consciousness doesn't exist this can be bad and i don't it it pains me a lot because i'm like well you know i'm teaching about consciousness and uh because that's what this theory is about but i don't really think that you can learn about consciousness from a second hand source and actually get it and not misinterpret it and kind of eventually do you realize your own existence so my advice to anybody listening to my theory or carl kristen's work or any other theory ctmu that is about consciousness your experience has to be the primary data set by which all other day data are compared and if they don't it doesn't uh align with what your personal experience is then disregard it don't think that your personal experience is wrong personal experience is right if you have to do this you absolutely have to you know i've said on here before that there i asked myself like first isaac asimov wrote a short story called last question and ends with let there be light it's it's interesting you should look it up it's about like a tech uh an ai and like basically creating universes and so i asked myself what is the first question and i found there actually is a first and the first question when you ask you say what is the first the answer is what is the first question it answers itself with itself you get to uh what is the second and it is what is the truest thing that i know and the second answer which is the answer to that question the answer to the second question is what is the first question it's the first question the first answer is what is the first second question is what is the truest thing that i know the second answer is i am and this this is like the fibonacci sequence one one two three it's like the first one repeats twice and then you get it kind of creates the second question and or the the ability to have the second question and then is answered uh as the third piece of their perspective or the fourth perspective but it is the uh it's like the second answer to the second question which would be i am you are the truest thing that you know everything starts with that that you exist i am and anything that invalidates that is false and it must be treated as false or you will you will lose yourself literally lose and uh you see this a lot with people who are in like politics these days where people have a meme called npcs non-player characters it's like just everybody's

becoming bots and they're not always bots they're bots in specific instances but they're increasingly bots and the reason why is because it's people who are trusting second-hand sources of information above their first-hand experience say oh well you know the everyone is is racist and sexist and this and and whatever you know you mean by that doesn't matter it means something negative and then you hear that again and again and again you've learned it in schools and that these people are oppressed and these people don't need uh these people men don't need women and women don't need men you know it's like if you start taking all this stuff again and again and again and again the second-hand information above your first-hand information which is like do women need men do men need women yeah your first-hand experience would absolutely tell you that you were birthed by a woman if you're a man then men need women you needed your mother your mother to give birth to you but if you start taking you know someone's word that like men don't need women and it's like or women don't need men it's like you're taking some exterior word from this and you're not thinking about your first hand experience um don't do that because the more you do that the more you will lose yourself but if you do that with consciousness you can totally lose yourself in like one or two you know um misconcluded ideas uh so i have to go here in a minute but uh i thank everybody for the support in the chat i'm going to answer these last things because uh someone had asked me about this in theory of everyone discord could you differentiate sentient non-sentient parts of reality yes non-sentient parts of reality would not be in this hierarchy of nested structures right here of like a hadron is an assembly of quarks an atom is an assembly of hadrons uh a molecule is an assembly of atoms cell is an assembly of molecules families and assembly of organisms etc so this nested helonic periodicity anything outside of it is uh non-sentient and this is each one is within the next an atom hadrons are inside of atoms atoms are inside of molecules molecules are inside of cells cells are inside of organisms organisms are inside of families families are inside of castes casts are inside of technology or you could say each one makes up the next hadrons make up atoms atoms make up molecules molecules make up cells cells make up organisms organisms make up families families make up casts casts make up technology and anything outside of that is going to be a nonsentience um okay is every split in the unity feeling something yes every object every sentient singularity within the singular sentient singularity of existence itself feels that's by definition that is that feeling is the finality of thought or information processing around the self so it's self aware is is a byproduct of self-awareness feeling so everyone feels kind of not everything not you know chairs do not feel but the molecules within it feel um like this does

not exist objectively it only exists subjectively this is a remote control for the lights it's made of molecules they do exist objectively and the atoms that make up those molecules also exist objectively because they're part of this periodic structure but you can't be like remote controls and get anywhere remote controls make up what you can see like remote controls are makeup made up of what plastic springs rubber glass you know like batteries that are you're just eventually going to get to rock bottom but it all of those are just subjective there isn't uh really an objective perspective to look at things at um so you have to go in that periodicity that i showed i don't have the time to get too into it i wanted to get more into it but this went a bit long um but i will talk to you more about it um and also this will all be in my paper that's coming soon is a story and i'm creating my head a separate cluster of awareness no only god's thoughts are sentient you are sentient thought but it's not even right to say thoughts are sentient because it's like that's not really true it's like you're a player in a thought but our concrete is the thoughts of the being above us so like god's thoughts or the universe's thoughts are experienced by us as concrete reality even though our thoughts are not experienced as by god as concrete reality or by us as concrete reality they are just internal states but because we are inside of a being above us in the hierarchy of nested periodicity of sentience anything in that is happening in the above us which is a mind in it of itself that's what it is a universal mind the thoughts of the universal mind are literally experienced by us as material reality it's higher their thoughts are higher than us in the hierarchy that doesn't mean you can't influence them it just means that like we experience them as hard reality you know you can influence hard reality but uh you know you can push a dresser across the room or whatever but like you're gonna can't just like think that dresser is not there and it's not there the thought in the mind that we're within is thinking that that dresser is there so it's there it's not a separate cluster of awareness there's only sentient beings and really and like they each have like various awarenesses but a thought is not a separate sentient being yeah the relationships between sentient beings can be the thoughts of the mind that we're within so think about your thoughts as a mind like imagine the universe is a mind and it's having thoughts and we are inside that mind that processes that are occurring within the mind that we are within are going to be experienced as hard reality to us because they are like you're inside of um how do you decide was an individual and what is not uh what i said earlier that periodicity is uh those composite structures are those are sentiences and uh so hadron may be an atom definitely molecule cell organism family cast and and technology and some half of they're really four beings or four sides of this periodicity each but this these eight periods can be split into four and four and

four of them maneuver in space via individuals and for the maneuver in time via lineages so an organism is maneuvering in space as an individual but it's lineage it's really part of a lineage structure is actually maneuvering in time via lineage of like chain a chain of individuals so uh and they're really the same thing you are not just one individual you are the tip of the sphere of a lineage that you know is behind you so it's a superposition structure but uh which one is an individual uh an individual lineage is also an individual it's kind of but it's also it's also a chain you are also an individual but you're also you know an assembly of cells so an individual is uh wherever being can be self-aware and meaning feeling it has feelings if it doesn't have feelings it's not an individual but if it has feelings like real feelings it is an individual so with that thank you everyone and uh would you describe god as god is the uh singular sentience that is existence itself existence is a sentient being all of existence yeah that is gone with that hit the like button please share this um i do have to go but uh i really appreciate everybody on here definitely check out the entire interview with kurt and um and carl fristen i'm going to watch the next uh um the other stream with carl fristen that was uh older but it it's like three and a half hours long or something like that and we'll see if there's more to talk about but his work is incredible to curt if kurt ever watches this or anyone else who is also investigating these theories of conscious that involve and you are the evidence of your own existence and do not believe anything that violates your personal experience even if i say it if i say something and it does not jive with what your personal experience is then don't believe it put it into a separate like basket of facts or not facts but of of you know bits of information that you've heard that may or may not be true the things that you put into the basket true must align with your personal and that is how you keep your sanity uh when you do this and end in anything your life is the best evidence you have of anything and uh the logical extrapolations that you can make from your own personal uh and beingness are what you should trust not what some second-hand source says in a book or even a person on youtube and then you know the extrapolations that you make from there you're going to go into nonsense land even for things as simple as like don't you know the world is sphere is that your experience i'm not saying you should say that it's it's flat in every instance but i'm just and i'm not definitely not what people would define as flat earther but i just think space is an emergent phenomenon guys like it's not fundamental so that's my position on it so we should just there's no when you're outside the space looking in on that space it's a sphere when you're inside of a space looking out it's a you're gonna experience plane so that's the reality like it's not even an argument but if you're either of those two perspectives where you're like the earth

is flat people are like the earth is a sphere we you know astronauts that have been outside of the earth and experienced it as a sphere like that guy who has experienced it as a sphere should never ever ever believe anybody who has uh who is saying it's just flat but he has two perspectives one is the perspective that this is flat and two the perspective that this is a sphere so he has two perspectives and he should believe both but if you've only been here then only one of those is one that you can truly trust as you know not a sphere but that doesn't mean you shouldn't believe that it is a sphere but it uh if you're outside it what it means is that you shouldn't let that thought override the thought or uh the experience that you have had firsthand that doesn't mean you can't believe that in some circumstance it it is uh congruent with or it's not mutually exclusive of your experience firsthand but it has to in some way eventually be reconciled with your experience firsthand if it's not then it's wrong and you have to treat it as such so with that thank you everyone please hit the like button please share this i hope this made sense um if not write in the comments below uh what you think didn't make sense or what you think did make sense and uh like i said with i will not be having the call on thursday uh i have a date but i will be back on tuesday and i believe next week should not be a problem either of having a call so we'll see depending on how quickly i get back maybe i'll jump on uh don't bet on it anyways with that peace guys and thank